



PLANTING OF COCONUT

Coconut is a perennial crop and expected to be productive for over 60 years if managed well. Therefore, it is very important to take correct decisions at the time of establishing a new coconut plantation. The essential factors which require the attention of growers at the time of establishing an either monoculture or multiple cropping plantations are, selection of planting materials adapted to the soil and the climate, appropriate planting distances and densities and correct planting and management of seedlings. Most coconut growers, including small holders are often not paid adequate attention on these crucial factors that determine the establishment of a highly productive coconut plantation.

A. Recommended planting materials

In view of long life span, coconut growers are advised to use the best planting materials for planting. The Coconut Research Institute currently recommends the following planting materials.



Figure 1: CRIC60 variety

1. Tall × Tall (CRIC60) variety

- ◆ This cultivar is developed by crossing within selected superior tall palms.
- ◆ This seed material is produced at the Isolated Seed Garden at Ambakelle and Seed Garden at Maduruoya.
- ◆ Annual production of CRIC60 seed nuts is about 1,000,000.
- ◆ Seedling certification tag is in dark blue color.
- ◆ Seedlings are available for growers from nurseries managed by both Coconut Research Institute and Coconut Cultivation Boards.
- ◆ Suitable for large scale coconut plantations.
- ◆ Recommended for all coconut growing areas.
- ◆ Under average management level, this planting material comes into flowering at 5-6 years of age.
- ◆ Annual nut production per hectare is about 12,000 - 15,000.

2. Dwarf × Tall (CRIC65) variety

- ◆ This hybrid is a cross between Sri Lankan Dwarf and Sri Lankan Tall varieties.
- ◆ Female parent of this hybrid is either Sri Lankan Green Dwarf or Sri Lankan Yellow Dwarf.
- ◆ Accordingly, there are two types of CRIC65 hybrids named CRIC65 green and CRIC65 yellow.
- ◆ CRIC65 yellow is showing a significant resistance to coconut mite but the nut yield is fairly less than CRIC65 green.
- ◆ These hybrids are produced at the Isolated Seed Garden at Ambakelle.
- ◆ Annual production of seed nuts is about 100,000.
- ◆ Seedlings are issued with a certification tag in dark green.
- ◆ Seedlings of CRIC65 are available for growers from nurseries managed by both Coconut Research Institute and Coconut Cultivation Board.
- ◆ Under average management conditions, this hybrid seedling come into bearing at 3-4 years of age.
- ◆ Annual nut production per hectare is about 20,000 - 22,000.



Figure 2: CRIC65 variety

3. Tall × San Ramon (CRISL98) variety

- ◆ CRISL98 is a cross between Sri Lankan Tall and Philippine San Ramón Tall variety.
- ◆ Produced at the Seed Garden at Pallama.
- ◆ Annual seed nut production is about 30,000
- ◆ Seedlings are certified with dark red sticker.
- ◆ Seedlings are issued to growers from nurseries managed by CRI.
- ◆ Suitable for large scale cultivations.
- ◆ Recommended for cultivation in all coconut growing areas of wet and intermediate zones.



Figure 3: CRISL98 variety

- ◆ Under normal management, seedlings come into bearing at 4-5 years of age.
- ◆ Annual nut production per hectare is about 12,000 - 15,000.
- ◆ CRISL98 is almost similar to CRIC60 in respect of adaptability to various climatic conditions and ability of nut production.
- ◆ The specific feature is ability of giving high copra yield.
- ◆ As a result of large nut size and thick kernel copra yield per nut is high as 280 -300 g.

4. Kapruwana (CRISL2004 / DG x SR)

- ◆ A hybrid between Sri Lankan Dwarf Green and Philippine San Ramon Tall variety.
- ◆ Produced by hand pollination at the Isolated Seed Garden, Ambakelle and the Seed garden, Pallama.



Figure 4: CRISL2004 variety

- ◆ Annual seed nut production is about 5,000.
- ◆ Seedlings are certified by an orange colour sticker.
- ◆ Seedlings are available for growers at nurseries managed by CRI in both Ambakelle and Pallama seed gardens.
- ◆ Under normal management condition seedlings come into bearing at 3-4 years after transplanting.
- ◆ Annual nut production per hectare is about 20,000 - 22,000.
- ◆ For mass production of seedlings, CRI established a seed garden at the Kiniyama Estate which belongs to the Chilaw Plantations. During the next five years, CRI is expected to supply seed nuts in large scale.

5. Kapsuwaya (CRISL2012 / DB x T)

- ◆ This hybrid was recommended to growers in 2012.
- ◆ Hybrid between Sri Lankan Brown Dwarf variety and Sri Lankan Tall variety.
- ◆ Seed nuts are produced by hand pollination at the Seed Garden at Pallama.
- ◆ Annual seed nut production is about 2,000.

- ◆ Seedlings are certified by a pink colour sticker.
- ◆ Seedlings are available for growers from the CRI nursery located at Seed Garden at Pallama.
- ◆ Under normal management conditions seedlings come into bearing at 3-4 years after transplanting.
- ◆ Annual nut production of this hybrid is about 20,000 - 22,000 per hectare.



Figure 5: CRISL2012 variety

6. Kapsetha (CRISL2013/DB x SR)

- ◆ This variety was recommended in 2012.
- ◆ A hybrid between Sri Lankan Dwarf Brown variety and Philippine San Ramon Tall variety.
- ◆ Seed nuts are produced by artificial hand pollination at the Seed Garden, Pallama.
- ◆ Seedlings are certified by a black colour sticker.
- ◆ Annual production of seed nuts is about 2,000.
- ◆ Seedlings are available for growers from the CRI nursery located at Seed Garden at Pallama
- ◆ Seedlings come into bearing at 3 - 4 years of age under normal management conditions.
- ◆ Annual nut yield of this hybrid is about 20,000 - 22,000 per hectare.



Figure 6: CRISL2013 variety

7. Moorock Tall

- ◆ This tall variety is selected from Moorock Estate located at Mawathagama.
- ◆ Recommended for cultivation in the wet and wet intermediate zones in Sri Lanka.
- ◆ Seedlings come into bearing at 5-6 years of age under normal management conditions

- ◆ Annual nut yield per hectare is about 11,000.
- ◆ Seedlings are certified with a white color sticker
- ◆ Seedlings are available for growers at nurseries managed by the Coconut Cultivation Board.

8. Plus Palm seedlings

- ◆ The existing coconut seed gardens in Sri Lanka are producing approximately 50% of the annual national demand of seed nuts. The deficit is collected from ordinary tall palms with high yield potential in high yielding estates. Seed nuts obtained from these selected palms are used to raise seedlings named as plus palms to be issued to growers.
- ◆ All characters are similar to CRIC60 variety.
- ◆ Recommended for cultivation in all coconut growing areas.
- ◆ Under normal management condition, nut production per hectare is about 8,000 - 9,000.
- ◆ Available for growers both as poly-bagged and bear rooted seedlings.
- ◆ Seedlings start bearing at the age of 5-6 years under normal management.
- ◆ Issued to growers from the nurseries managed by the Coconut Cultivation Board.



Figure 7: Stickers used to certify coconut seedlings

B. Planting densities

In coconut planting, the optimum density of coconut is recommended as 158 palms / ha (64 palms / ac).

2. Inter - Cropping with coconut

01. Recommended spacings for coconut monoculture or mixed cultures with annual or biannual intercrops.

Square method

Spacing		No. of seedlings	
Meters	Feet	Per ha	Per ac
8.0 x 8.0	26 x 26	158	64

- ❖ Refer Advisory Circular C for more details about annual and biannual intercrops .

02. Recommended spacings for coconut and perennial intercropping systems.

Wide row method

Spacing		No of seedlings	
Meters	Feet	Per ha	Per ac
7.8 x 9.6	26 x 32	128	52

- ❖ Refer Advisory Circular C for more details about perennial intercrops .

03. Recommended spacings for coconut and tea/sugarcane intercropping systems.

Special wide row method

Spacing		No of seedlings	
Meters	Feet	Per ha	Per ac
7.8 x 12	26 x 40	102	42

- ❖ Priority is given for the cultivation of tea/sugarcane, but coconut is cultivated to get additional income.

C. Replanting and under planting

- ❖ When the coconut plantation is senile and nut production becomes low, growers should decide to re-plant their plantations without delaying. The following points are to be considered in re-planting.



Figure 8: A new plantation

- ◆ Age of the plantation - Majority of bearing palms are over 60 years of age.
- ◆ Nut production - Under general conditions, annual nut production per acre is less than 1000 nuts.

- ◆ Height of palms.
- ◆ Number of palms - less than 50 palms per acre.

The following methods are recommended in establishing new plantations instead of senile plantations.

Replanting : Planting new seedlings in the same land after complete removal of old palms.

Under planting: Planting new seedlings in between the exiting rows of old plantation and gradually remove them within a 5 - 6 year period after planting of seedlings.

- ◆ Based on experimental data, replanting has proved superior to under planting in terms of early bearing, growth and nut production.
- ◆ Replanting is recommended for large scale plantations with high level of management.
- ◆ But for small holders, under planting is recommended due to financial constraints in replacing senile palms.



Figure 9: An under plantation

❖ Replanting

- ◆ Old plantation should be removed completely. If the removal of root bole is difficult, cut stems at the ground level and cover with soil sufficiently or apply coal tar on the cut surface to prevent Black Beetle breeding.
- ◆ Select the appropriate planting system and the number of seedlings per unit area and mark the planting holes.

❖ Under planting

- ◆ Line the area and mark the planting holes.
- ◆ The new line should be based along the middle of the existing coconut square (to reduce the completion between old and new palms for

sunlight, water and nutrients). Cut the planting hole in the center of the old square (figure 9).

- ◆ Remove all palms falling within a minimum distance of 8 feet from the newly marked planting points (20% generally).
- ◆ Remove the remaining old stand gradually in stages based on the distance from the seedling to ensure adequate sunlight availability, as indicated below.
 - 1st year after planting - 15%
 - 2nd year after planting - 15%
 - 3rd year after planting - 20%
 - 4th year after planting - 20%
 - 5th year after planting - remaining palms
- ◆ If annual removal is difficult, thinning can be carried out in the following manner.
 - 2nd year after planting - 30%
 - 4th year after planting - 40%
 - 6th year after planting - remaining all palms
- ◆ Systematic removal of the old stand is of utmost importance for the growth and flowering of the new planting.

Lining

To mark the planting holes at the correct distance and density one of the following methods could be adopted.

1. Square planting (8 m spacing)

Step 01: Mark the baseline

- Make the baseline using a rope about 4 m from the boundary of the land.

Step 02: Mark a perpendicular line

- Select point A on the first peg of the baseline.
- Mark a distance (B) 20 m away from A on the baseline.

- Select point C exactly 15 m away from A and 25 m away from B.
- Mark the perpendicular line by joining A and C.
- This is the reference line to adjust the planting points (figure 10).

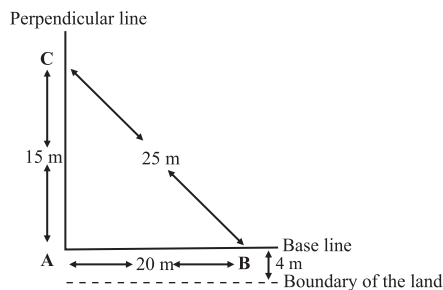


Figure 10: Marking of perpendicular line

Step 03 : Mark planting points at 8 m intervals on the baseline and reference line (rope).

- Use a 16 m tape with a peg at the center.
- Place two free ends of the tape at points B and C on base and reference line respectively
- The point at the peg D is the fourth planting point of the square system (figure 11).
- This has to be repeated for pegging other planting points.
- At every fifth rows check and adjust the planting points with the reference line.

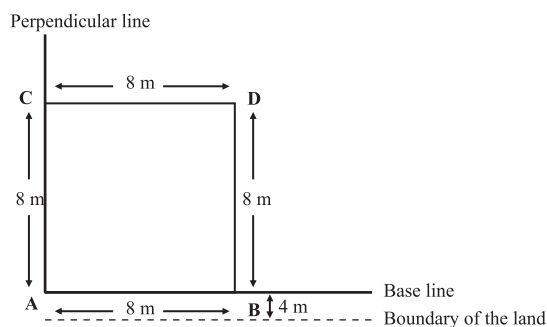


Figure 11: Marking holes with square method

D. How to plant a coconut seedling correctly in the field

❖ Place to be planted

- ◆ Select a suitable place for planting prior to cutting a hole.
- ◆ Sandy loam soil with good drainage and full sunlight availability is regarded as the best place for planting a seedling.
- ◆ As the available spaces in a home garden is limited, it is important to select a suitable place for planting with full sunlight considering the landscaping beauty of a home garden and a building.
- ◆ If a coconut seedling is planted in a shady place, good growth could not be achieved even with the correct application of fertilizer and water.
- ◆ In urban home gardens, most limiting factor for seedling growth is non availability of full sunlight.

❖ Correct method of planting

- ◆ The size of the planting hole is determined by the type of soil. In sandy and loamy soils the size of the planting hole should be 3' x 3' x 3' feet and in gravel and cabook soils the size of the hole should be 4' x 4' x 4' feet.
- ◆ In lands with hard soils, the use of back-hoe machines is economical to cut planting holes.
- ◆ When the planting hole is cut, place the top soil layer (up to one foot depth) a side and the rest of the soil on the other side.
- ◆ As the top soil layer is rich in nutrients use this soil first to fill the planting hole. Subsoil should be improved as described below before using it for filling holes as its nutrient levels are low.
- ◆ Place two layers of coconut husks or coir dust at the bottom of the hole.
- ◆ Firstly place one layer of husks and cover them with top soil. Then place the second layer of husks as in the first layer and cover that too with top soil.
- ◆ When coconut husks are used place them with fiber side upwards. Each planting hole requires about 30 - 35 husks.

- ◆ Coconut husks absorb water nearly 6 -7 times of its weight during the rainy periods and gradually releases water during the dry periods which is absorbed by the roots of palms.
- ◆ To fill the rest of the planning hole, subsoil is mixed with the following ingredients.
- ◆ Organic materials such as 5 kg of wood ash or 10 kg of cow dung or 5 kg of goat dung, with 1 kg of dolomite and 1 kg of Young Palm Mixture (YPM), the composition of which is indicated in the Table 1. Mix all materials with subsoil and fill the planting hole (figure 12).
- ◆ The above mixture should be filled up to about 20 cm below the ground level (figure 13).



Figure 12: Materials used to plant a coconut seedling

Table 1 : Young Palm Mixture

Wet and intermediate zones		Dry zone	
Fertilizer	Amount (g)	Fertilizer	Amount (g)
Urea	250	Urea	250
Eppawela Rock Phosphate	750	Triple Super Phosphate	350
Muriate of Potash	250	Muriate of Potash	250

- ❖ In an area where water logging is a problem, fill the hole up to the ground level.

- ◆ If water is collected around seedlings during the rainy period, elevate the planting point by collecting more soil.
- ◆ Avoid planting seedlings deep in the soil to prevent from rotting the collar region due to fungal attack.



Figure 13: LS of a coconut seedling hole

- ◆ In order to have a higher yielding plantation, select only certified seedlings in poly-bags for planting.
- ◆ For planting in home gardens, select hybrid seedlings of CRIC65, Kapsuwaya, Kapsetha, and Kapruwana which come into bearing early. For

large scale plantations, select hybrid varieties such as CRIC60, CRISL98 (T x SR) and Plus Palms and Morock Tall.

- ◆ When poly-bagged seedlings are used for field planting, first cut and remove the bottom of the poly bag with a blade.
- ◆ Then place the bottomless poly bagged seedling in the center of the above prepared hole after removing the soil to accommodate the poly bag in the soil mixture.
- ◆ Finally lift and remove the poly bag carefully without disturbing the soil in the poly bag and roots of the seedling.
- ◆ In areas where Black Beetle damage is high, poly bag may be lifted half way and tied it up to the base of the seedling to cover the collar region of seedling (figure 14) to protect seedlings from Back Beetle at early stage.



Figure 14: Correct method of attaching polybag to the coconut seedling

- ◆ If poly bag is tied to the collar region of seedling, made a few holes on the poly bag to facilitate aeration.
- ◆ This poly bag should be removed after 3 - 4 months from planting.
- ◆ However, in poorly drained soils, poly bag tying around the base may increase the incidents of collar rot during rainy season. In such situations remove the poly bag completely from the collar region to facilitate the direct exposure to sunlight.
- ◆ After planting seedlings in well-drained soils, make a basin like structure around seedlings to hold more water (figure 15).
- ◆ But in poorly drained soils lift the planting point by heaping up soil.
- ◆ To conserve soil moisture, it is important to place coconut husks, pieces of dried fronds and dried pasture around the seedlings as a mulch.

- ◆ In poorly drained areas, planting of seedlings should commence at the latter part of rainy period.
- ◆ To protect seedlings from termite attack, 4 ml of Marshal 20% diluted with 1 liter of water and immerse the nut of the seedling for 3 -5 minutes in the case of planting bare-rooted seedlings. After planting the poly-bagged seedlings in the field, 5 liter of the same dilution of the pesticide should be drenched well around seedlings.

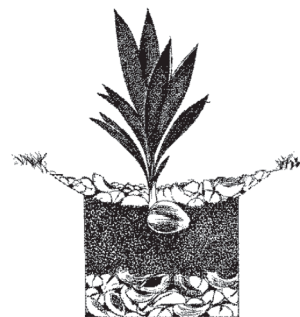


Figure15: Correctly planted coconut seedling

❖ **Time of planting**

Seedlings should be planted with the onset of rains. But in water logging areas, planting should be carried out towards the end of rainy season. Planting with the rains enables seedlings for root development before the dry season. Planting should be planned in Yala season for the wet zone and Maha season for the dry zone. It is advisable to dig the holes and fill them during the inter-monsoonal periods. Rows of the coconut should be in east-west direction when square or rectangular planting is adopted. Irrigating the seedlings during dry season and soil moisture conservation practices will help proper growth of seedlings.

❖ **Infillings**

To maintain a uniform plantation, at the end of the 1st year after planting, all weak seedlings should be removed and replace them with healthy seedlings is essential. It is recommended that infilling should be carried out up to 15 years from the date of planting. Infilling is not possible after 15 years.

Coconut nurseries belong to the Coconut Cultivation Board where coconut seedlings are available for growers.

Name of the nursery	Address of the nursery	Name of the nursery	Address of the nursery
1. Kiridiwela	Coconut nursery, Kiridiwela. 033-2267704	2. Wilpotha	Coconut nursery, Wilpotha, Baththuluoya 032-5671077
3. Walpita	Coconut nursery, Walpita. 033-2272559	4. Wennoruwa	Coconut nursery, Muthugalawatta, Wennoruwa 037-2249836
5. Polgahawela	Coconut nursery, Serapis watta, Panaliya, Polgahawela 037-2244367	6. Attawilluwa	Coconut nursery, Kurunegala Road, Attawiluwa, Puttalam 032-5670793
7. Hadapanagala	Coconut nursery, Hadapanagal Weherayaya 055-2274665	8. Labuduwa	Coconut nursery, Akmimana, Galle 091-2234501
9. Eraminigolla	Coconut nursery, Eraminigolla, Hiriwadunna 035-2263006	10. Medamulana	Coconut nursery, Medamulana. 047-2246359
11. Milambaweli	Coconut nursery, Milambaweli, Nenkala, 065-2240924	12. Diyadora	Coconut nursery, Pahaladiyadora 037-2282537
13. Dodamgaslanda	Coconut nursery, Balawatta Watta, Dodamgaslanda. 037-2252625	14. Welipitiya	Coconut nursery, Welipitiya, Nanapurawa, Bibila. 055-2267720
15. Nikaweratiya	Coconut nursery, Nikaweratiya. 037-2260619	16. Weligama	Coconut nursery, Pathegama, Weligama 041-2252336
17. Bingiriya	Coconut nursery, Presannagama, Bingiriya 032-2246425	18. Baddegama	Coconut nursery, Pedaruswatta, Halpathota, Baddegama. 091-2292745
19. Kandetiya	Coconut nursery, Kandetiya, Makandura, Gonawilla. 031-2299386	20. Korosduwa	Coconut nursery, Korosduwa Watta, Wadduwa. 038-2295126
21. Kirimetiya	Coconut nursery, Kirimetiya, Lunuwila, 031-2257504	22. Pallakele	Coconut nursery, Pallakela, Kunadasala 081-5040595
23. Ibbagamuwa	Coconut nursery, Ibbagamuwa, Thorayaya. 037-2235776	24. Kalawewa	Coconut nursery, Kalawewa, Wijithapura 025-2264194
25. Uhana	Coconut nursery, Palam Kadaura, Uhana. 063-2250119	26. Ulhitiya	Coconut nursery, Ulhitiya Watta, Giradurukotte 027-3279230
27. Prakrama Samudraya	Coconut nursery, Richad Aluwihare Udyanaya, Prakrama Sumudraya 027-2224304	28. Mahayaya	Coconut nursery, Mahayaya 031-2299938
29. Wariyapola	Coconut nursery, Wariyapola. 037-2267691	30. Embilipitiya	Coconut nursery, Embilipitiya. 047-2230569
31. Maththamagoda	Coconut nursery, Maththamagoda. Kannathota. 035-2289105	32. Hettipola	Coconut nursery, Hettipola 037-2291332

Genetics and Plant Breeding Division
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